

Nonnegative and positive factorizations

Rational factors for rational completely positive boundary matrices

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Extreme because rational certificates on arbitrary boundary faces require new arithmetic control beyond interior geometry; community importance is exact certification in completely positive optimization.

Area: exact certificates for nonnegative symmetric factorization

Context and notation

A symmetric matrix A is **completely positive** if $A = BB^T$ for an entrywise nonnegative real matrix B with finitely many columns. Let $\mathcal{CP}_n$ be the cone of these matrices of order n . Its **cp-rank**, $\text{cpr}(A)$, is the minimum number of columns in such a factor, with $\text{cpr}(0) = 0$. Boundaries and interiors use the usual Euclidean topology on the vector space of real symmetric matrices.

Problem statement

For every integer $n \geq 5$ and every

$$A \in \mathbb{Q}^{n \times n} \cap \partial \mathcal{CP}_n,$$

does there exist a finite integer $m \geq 1$ and a matrix $B \in \mathbb{Q}_{\geq 0}^{n \times m}$ such that

$$A = BB^T?$$

The width m is unrestricted. In particular, it need not equal the real cp-rank or the order of A . This is an existence question for an exact rational certificate of complete positivity.

Rational matrices in the interior of $\mathcal{CP}_n$ have rational completely positive factors. Several boundary classes are also known, including matrices of rank at most two. The question concerns the remaining boundary matrices.

References

Abraham Berman and Naomi Shaked-Monderer, *Completely Positive Matrices over Sets*, Communications in Optimization Theory (2025), article 23, §2, Question 2.1 and the explicit boundary Open Problem, p. 2. Mathieu Dutour Sikirić, Achill Schürmann, and Frank Vallentin, *Rational factorizations of completely positive matrices*, Linear Algebra and its Applications **523** (2017), 46–51, Theorem 1.1. Max Pfeffer and José Alejandro Samper, *The Cone of 5×5 Completely Positive Matrices*, Discrete & Computational Geometry **71** (2024), 442–466, §6, Problem 6.3, discusses the order-five boundary.

Status check — 2026-09-10

Rechecked [Berman–Shaked-Monderer, §2](#) and [Oertel–Schürmann v2, §6.1.1](#), and searched for a 2026 boundary resolution. Rank-at-most-two and other specified boundary classes have rational factors, but the 2025 paper still poses the general boundary problem. The 2026 interior theorem does not resolve it. No full boundary construction or counterexample was located; unrestricted factor width is essential.